

Dimitrios Panagiotis Emmanouilidis

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Software developer — AI engineering: local LLM inference, retrieval, evaluation, and safety gating for model-driven actions.

TECHNICAL SKILLS

Languages: Python, C++, Kotlin

Platforms & Tooling: Android (NDK, JNI, Gradle), Linux, Windows, Git, SQLite, GPU/VRAM budgeting, cross-platform toolchain debugging

AI & Machine Learning: Ollama, llama.cpp/GGUF, local LLM inference, quantised models, hybrid retrieval (dense + lexical), JSON Schema validation, deterministic verification of model output, evaluation harness design

TECHNICAL PROJECTS

casefile — Evidence-Constrained Case Generation with a Local LLM

Sole Author · Public · Sep 2026 · github.com/dpemmanouilidis/casefile

- Self-hosted on-chain investigation pipeline in four stages: ingest, enrichment and flow tracing, a deterministic rule gate, and narration by a local LLM. Rules decide what is flagged; the model only restates established facts and never contributes one of its own.
- Seven verification checks run on every generated narrative before it is shown; four are exact and reject outright (cited hashes and addresses must exist in the evidence, the stated verdict must match the gate's, low confidence must be declared). Failure produces an explicitly marked template, never unverified prose.
- Measured 16 stored verdicts across four model configurations: qwen3.5:9b passed every check on the first attempt at Q4_K_M and Q8_0; qwen2.5:3b managed 12%. Citation precision 100% on every row. Two prompt constraints for an advisory check cost one configuration 25 points of first-attempt pass rate on unrelated checks — reverted and moved into verification.
- Verdict replay caught six stored verdicts that were silently wrong (an obsolete rule's precedence masking real findings). Fails closed throughout: a rule that cannot be evaluated returns unknown, never a default pass. 143-test offline suite. Python, SQLite, Ollama. MIT.

actuation-runner — LLM Action-Payload Failure Benchmark

Sole Author · Public · Aug 2026 · github.com/dpemmanouilidis/actuation-runner

- Open-source benchmark measuring how often LLM-generated action payloads fail before reaching an actuator, classified across eight outcome categories (one pass, seven distinct failure modes).
- Two independent 100-trial runs against a local quantised 9B model: 89/100 and 92/100 valid. Every failure across all 200 trials was planner token-ceiling truncation — a one-to-one match with the model's length stop reason. The run-to-run spread is itself the finding: either run alone would have read as a precise figure.
- 45-test offline suite, including tests that invoke the CLI as a subprocess so the real invocation path is exercised end to end. Three defects caught before the measured run by reading source and showing computed inputs prior to execution, among them a CLI entry point that printed results while writing nothing to disk. Python, jsonschema, Ollama. Apache 2.0.

Offline Local Inference & RAG Application

Lead Developer · In Development · Private repo, walkthrough available on request

- Android app running a quantised LLM entirely on-device, with no network at query time; llama.cpp/GGUF core over FFI behind a native Jetpack Compose UI.
- Hybrid retrieval fusing 384-dim MiniLM dense vectors with a character-trigram BM25 index. Answers carry passage citations and refuse when the corpus does not contain the answer.
- Footprint budgeted against Google Play asset delivery: a 46 MB embedding model beside a 2B-class Q4_K_M LLM at about 1.3 GB, inside the 1.5 GB per-pack limit. Verified on a Samsung Galaxy S24.
- Built the retrieval eval harness that gates release, including a hand-curated catastrophic-miss list that blocks a corpus build from shipping regardless of average recall. Kotlin, C++, llama.cpp, all-MiniLM-L6-v2, Room/SQLite, Python build pipeline.

Contextual Synthesis and Constrained Action: Evaluating a Two-Agent LLM System for Personalized Vehicle Adaptation

BSc Dissertation · Jan–May 2026

- Two-stage LLM pipeline (profiler feeding planner) behind a non-LLM JSON Schema safety gate, tested in the CARLA simulator; 4-bit quantised 9B model via Ollama in a 12 GB VRAM budget shared with the simulator.
- Measured end-to-end latency and reported it as prohibitive for real-time control on that hardware. Found an idle simulator process halved local inference throughput (roughly 51 tok/s against 93 tok/s) with GPU utilisation still reporting 100%.

EDUCATION

BSc (Hons) Computing (Application Development)

New York College / University of Greater Manchester — Thessaloniki, Greece · Graduated July 2026